

KRISHNA MRIDHA

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EDUCATION

- **Case Western Reserve University** Cleveland, OH, USA
Ph.D. in Biomedical and Health Informatics — CGPA: 3.75/4.0 *August 2023 – Present*
- **Bachelor of Technology in Computer Engineering** Marwadi University, Gujarat, India
CGPA: 9.02/10 (Top 5%) *June 2019 - May 2023*
- **Diploma in Computer Technology** SPPI, Shariatpur, Bangladesh
CGPA: 3.70/4.0 *December 2013 - December 2017*
(1st position out of 400 students)

PROGRAMMING SKILLS

- **Languages:** Python, R, C, Java
- **Technologies and Frameworks:** TensorFlow/Keras, Flask, Django, OpenCv, Scikit-learn
- **Database:** MySQL, NoSQLite, Oracle

RESEARCH INTEREST

Biomedical Data Science; Medical Imaging (X-ray, CT, MRI, Histopathology); Multimodal Deep Learning; Explainable and Trustworthy AI; Clinical Informatics; Machine Learning for Healthcare; Physiological Signal Processing (EEG, ECG, EMG); Digital Health and Remote Monitoring; AI for Mental and Neurological Disorders.

JOB EXPERIENCE

- **Saic Institute of Medical Technology** Dhaka, Bangladesh
Junior Instructor (ICT) – Full-Time *August 2017 – August 2019*
 - Delivered ICT training with emphasis on digital tools for healthcare data entry, medical record management, and reporting systems.
 - Taught medical students the use of MS Excel and Access for organizing patient information and generating clinical reports.
 - Supported the integration of digital systems within the institute's health training programs to improve record-keeping and administrative workflows.
 - Advised students on the use of health informatics tools to improve operational efficiency in clinical and hospital environments.

RESEARCH EXPERIENCE

- **Cleveland Clinic** Cleveland, Ohio, USA
Predoctoral Research Fellow, AI and Informatics in Healthcare Lab *June 2026 – Present*
 - Conducting research on AI-driven diagnostic systems in cardiology under Dr. David Chen
 - Developing uncertainty-aware and explainable deep learning systems for cardiovascular imaging
 - Collaborating with clinical faculty at the Heart, Vascular and Thoracic Institute
- **Case Western Reserve University** Cleveland, Ohio, USA
Graduate Researcher (PhD Student) *Aug 2023 – May 2026*
 - Developed explainable AI tools for clinical diagnostics and risk prediction

- Conducted research in health informatics, medical imaging, and electronic health records (EHR)
- Led multiple research projects in chronic disease modeling and AI for healthcare

- **Marwadi University** Rajkot, Gujarat, India
Undergraduate Research Assistant *February 2021 – May 2023*
 - Conducted research in AI/ML for medical imaging applications, focusing on disease classification and segmentation under the supervision of Dr. Madhu Shukla.
 - Applied computational data science techniques for health-related data analysis and model development.
 - Explored neural signal processing and deep learning frameworks in computational neuroscience projects.

INTERNSHIP EXPERIENCE

- **REGex Software Services** **(May 2021 – August 2021)** Tamil Nadu, India
 Paid internship involving analysis and modeling of real-time employee loan data. Developed a loan approval prediction model using Machine Learning algorithms.
- **Pianalytix** **(December 2020 – January 2021)** Hyderabad, India
 Paid internship as a Machine Learning research intern. Developed models primarily using the U-Net algorithm for medical image analysis.
- **TechCiti Technologies** **(November 2020 – January 2021)** Karnataka, India
 Unpaid learning internship focused on Stock Market Prediction using Machine Learning techniques. Utilized real-time data from more than ten companies for predictive analysis.

VOLUNTEER EXPERIENCE

- **IEEEXtreme** **(June 2021 – July 2022)** Gujarat, India
 Served as Campus Ambassador, promoting the IEEEXtreme Programming Competition to schools, universities, and IEEE Student Divisions. Assisted IEEE Student Branches in organizing competitions and guided non-IEEE students on participation processes.
- **Pianalytix** **(December 2020 – January 2021)** Hyderabad, India
 Acted as Campus Ambassador to enhance program awareness through promotional activities such as posters, flyers, social media outreach, information sessions, and collaboration with university departments and student organizations.

PROJECTS

- **Alarm based Driver Drowsiness Detection** A smart application for smart cities that can help to identify the drowsiness of driver [\[project\]](#)
- **Breast Cancer Prediction Model** A web based application developed using Machine Learning-Heroku-Flask to predict the Breast Cancer based on the Lab Report [\[project\]](#)
- **Real-time Smart Face Mask Recognition** It is one of the best Real-time applications that can recognize the people who are not wearing the mask developed by Convolutional Neural Network and OpenCv. [\[project\]](#)
- **Intelligent based Phishing Website Detection** A web based application developed using Artificial Neural Network Algorithm-Heroku-Flask-Bootstrap to identify the PHISHING URL. [\[project\]](#)

- **Automated estimating the approval of a bank's client loan.** A web based application developed using Machine Learning-Heroku-Flask-Bootstrap to predict the approval of Bank's loan based on the customer previous record [\[project\]](#)
- **Using Machine Learning In Predicting Diabetes.** A web based application developed using Machine Learning-Heroku-Stremlit to predict the Diabetes [\[project\]](#)

JOURNAL PUBLICATIONS

- **Krishna Mridha , Madhu Shukla, "Image Resolution Controls Calibration and Reliability in Deep Learning for Medical Image Classificationn"** Computers in Biology and Medicine [\[Under Review\]](#)
- **Krishna Mridha , Ajoy Chandra Kur, Trinoy Saha, Madhu Shukla"SmartHeart: Explainable Machine Learning for Cardiovascular Risk Prediction"** Computers in Biology and Medicine [\[DOI\]](#)
- **Krishna Mridha , Madhu Shukla, Biswaranjan Acharya, Vassilis C. Gerogiannis, Andreas Kanavos, Masrur Ahsan Priyok, Rayhan Hussain Razu "Implementing a Heart Disease Prediction Model with Explainable Machine Learning Techniques"** SN Computer Scieene [\[DOI\]](#)
- **Krishna Mridha , Masrur Ahsan Priyok, Madhu Shukla"Unraveling the Intricacies of EEG Seizure Detection: A Comprehensive Exploration of Machine Learning Model Performance, Interpretability, and Clinical Insights"** Multimedia Tools and Application [\[DOI\]](#)
- **Krishna Mridha ,Md Mezbah Uddin, Jungpil Shin, Susan Khadka, MF Mridha, "An Interpretable Skin Cancer Classification using Optimized Convolutional Neural Network for a smart Healthcare System"** [\[IEEE ACCESS - DOI\]](#)
- **Krishna Mridha ,Sandesh Ghimire, Jungpil Shin, Anmol Aran, Md Mezbah Uddin, MF Mridha, "Automated Stroke Prediction Using Machine Learning: An Explainable and Exploratory Study with a Web Application for Early Intervention"** [\[IEEE ACCESS - DOI\]](#)
- **Krishna Mridha ,Sourav Simanta, Milan Limbu, "U-Net for Medical Imaging: A Novel Approach for Brain Tumor Segmentation"** [\[Global Journal of Innovation and Emerging Technology - DOI\]](#)

CONFERENCE PUBLICATIONS

- **Krishna Mridha, K.C. Santosh Pareto-Descent Guarantees for Adaptive Multi-Task Optimization,** Efficient Deep Learning for Computer Vision (ECV) Workshop at CVPR 2026 [\[Under Review\]](#)
- **Krishna Mridha, Kc Santosh, "Causally Invariant and Uncertainty-Aware Neural ODEs for Retinal Disease Diagnosis",** Medical Image Computing and Computer Assisted Intervention (MICCAI 2026) [\[Under Review\]](#)
- **Krishna Mridha, Humayera Islam ,"Cross-Attention Enables Context-Aware Multimodal Skin Lesion Diagnosis",** American Medical Informatics Association (AMIA) 2026 Annual Symposium [\[Under Review\]](#)
- **Krishna Mridha, Ming Wang, Lijun Zhang, "Building Trust in Clinical AI: A Web-Based Explainable Decision Support System for Chronic Kidney Disease ",** American Medical Informatics Association (AMIA) 2025 Informatics Summit [\[DOI\]](#)
- **Krishna Mridha, Ming Wang, Lijun Zhang, "AI-Driven Diagnostics in Ophthalmology: Tailored Deep Learning Models for Diabetic Retinopathy with XAI Insights",** 16th International Conference on Bioinformatics and Computational Biology (BICOB-2024) [\[DOI\]](#)
- **Krishna Mridha, Tasnim Sarker, Rawn-oquz Zaman, Madhu Shukla "Emotion Recognition: A New Tool for Healthcare Using Deep Learning Algorithms",** 4th International Conference on Electrical and Electronics Engineering (ICEEE 2024) [\[DOI\]](#)
- **Krishna Mridha, Tasnim Sarker, Suborno Deb Bappon, Shahriar Mahmud Sabuj "Attention U-Net: A Deep Learning Approach for Breast Cancer Segmentation",** IEEE International Conference on Quantum Technologies, Communications, Computing, Hardware and Embedded Systems Security (iQ-CCHES 2024) [\[DOI\]](#)

- **Krishna Mridha**, MD. Mezbah Uddin, Ebisa Leta Desisa, Asefa Teganu Feyisa, Segni Worku Guta, Wolde Assefa Debele”**EEG-based Epilepsy Seizure Classification Using Explainable Machine Learning Algorithms**”, Fifth International Conference on Advances in Electronics, Computers and Communications (ICAEECC 2023) [\[DOI\]](#)
- **Krishna Mridha**, Suborno Deb Bappon, Shahriar Mahmud Sabuj, Tasnim Sarker ”**Explainable Machine Learning for Drug Classification**”, 4th International Conference on Electrical and Electronics Engineering (ICEEE 2024) [\[DOI\]](#)
- **Krishna Mridha**, Ajoy Chandra Kuri, Trinoy Saha, Nancy Jadeja, Madhu Shukla and Biwaranjan Acharya ”**Toward Explainable Cardiovascular Disease Diagnosis: A Machine Learning Approach**”, International Conference on Data Analytics and Insights (ICDAI-2023) [\[DOI\]](#)
- **Krishna Mridha**, Trinoy Saha, Ajoy Chandra Kuri, Apu Chandra Barman, Dharmdevsinh Jadeja ”**Implementing an Alarm Based Driver Drowsiness Detection System for Traffic Safety Using Neural Network**”, 2023 IEEE Renewable Energy and Sustainable E-Mobility Conference (RESEM) [\[DOI\]](#)
- **Krishna Mridha**, A. C. Barman, S. Biswas, S. Sarkar, S. Biswas and M. A. Priyok ”**Accuracy and Interpretability: Developing a Computer-Aided Diagnosis System for Pneumonia Detection in Chest X-Ray Images**”, 2023 International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE) [\[DOI\]](#)
- **Krishna Mridha**, Fistum Getachew Tola , Ibrahim Khalil, Sheikh Md. Jamiul Jakir, Pieboji Noubissie Wilfried ”**Explainable Deep Learning for Coffee Leaf Disease Classification in Smart Agriculture: A Visual Approach**”, 2023 International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE) [\[DOI\]](#)
- **Krishna Mridha**, Meghla Monir Shorna, Nazmul Arefin, Ananya Ritu, MD Minhazul Alam Chowdhury, MD Iftekhar Islam ”**DBNet: Detect Diabetic Retinopathy to Stop Blindness Before it’s Too Late**”, International Conference On Reliability, Infocom Technologies And Optimization (Icrito’2022) [\[DOI\]](#)
- **Krishna Mridha**, Ananya Ritu, Md Minhazul Alam Chowdhury, Nazmul Arefin ”**ML-MT: A Study of e-Health Application Framework by Machine Learning Techniques**”, 2022 IEEE 4th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA) [\[DOI\]](#)
- **Krishna Mridha**, ”**Bioactivity classification of SARS-CoV-2 Proteinase using Machine Learning Approaches**”, International Conference On Reliability, Infocom Technologies And Optimization (Icrito’2022) [\[DOI\]](#)
- **Krishna Mridha** MD Iftekhar Islam, Shamin Ashfaq, Masrur Ahsan Priyok, Dipayan Barua ”**Deep Learning in Lung and Colon Cancer classifications**”, 2022 International Conference on Advances in Computing, Communication and Materials (ICACCM) [\[DOI\]](#)
- **Krishna Mridha** MD Iftekhar Islam, Meghla Monir Shorna, Masrur Ahsan Priyok ”**ML-DP: A Smart Emotion Detection System for Disabled Person to Develop a Smart City**”, 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) [\[DOI\]](#)
- Ekanta Saha, Runa Saha, **Krishna Mridha**, ”**Short-Term Electricity Consumption Forecasting: Time-Series Approaches**”, 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) [\[DOI\]](#)
- **Krishna Mridha** ,Dipayan Barua, Meghla Monir Shorna, Hasan Nouman Nouman, Md Hasanul Kabir, Ajay Vikram Singh, ”**Credit Approval Decision using Machine Learning Algorithms**” 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions)(ICRITO) [\[DOI\]](#)
- **Krishna Mridha** ,Raj Panjwani, Madhu Shukla, ”**Face Mask Alert Detection System For Preventing the Spread of COVID-19**” International Conference on Control, Automation, Power and Singnal Processing (CAPS - 2022) [\[DOI\]](#)
- **Krishna Mridha** , ”**Early Prediction of Breast Cancer by using Artificial Neural Network and Machine Learning Techniques**”, 10th International Conference on Communication Systems and Network Technologies (CSNT),2021 [\[DOI\]](#)

- **Krishna Mridha** , Nabhan Tawjih Yousef, ”**Study and Analysis of Implementing a Smart Attendance Management System based on Face Recognition Tecqnique using OpenCV and Machine Learning**”, 10th International Conference on Communication Systems and Network Technologies (CSNT),2021 [\[DOI\]](#)
- **Krishna Mridha** , Dinesh Kumar, Madhu Shukla, Mahrishi Jani, ”**Temporal Features and Machine learning Approaches to Study Brain Activity with EEG and ECG**”, International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 2021 [\[DOI\]](#)
- Ronak Doshi, **Krishna Mridha**, Dinesh Kumar, Amit Ved, ”**Medium and Short Term Energy Forecasting using LSTM Neural Network Method for Gujarat State**”, Asian Conference at Innovation in Technology, 2021 [\[DOI\]](#)
- **Krishna Mridha** , Rabindra Nath Shaw, ”**Intelligent based Waste Management Awareness Developed by Transfer Learning**”, International Conference On Computing, Power And Communication Technologies (Gucon), 2021 [\[DOI\]](#)
- **Krishna Mridha** , Md Jahid Hasan, Rabindra Nath Shaw, ”**Phishing URL Classification Analysis Using ANN Algorithm**”, International Conference On Computing, Power And Communication Technologies (Gucon), 2021 [\[DOI\]](#)
- Raj Chokrobarty, **Krishna Mridha**, Rabindra Nath Shaw, ”**Study and Prediction Analysis of the Employee Turnover using Machine Learning Approaches**”, International Conference On Computing, Power And Communication Technologies (Gucon), 2021 [\[DOI\]](#)
- **Krishna Mridha** , Akshay Ranpariya, Smit Kumbhani, ”**Plant Disease Detection using Web Application by Neural Network**” , 2021 IEEE 6th International Conference on Computing, Communication and Automation (ICCCA) [\[DOI\]](#)
- **Krishna Mridha** , Smit Kumbhani, Suman Jha, Dhara Joshi, ”**Deep learning algorithms are used to automatically detection invasive ducal carcinoma in whole slide images**” , 6th IEEE International Conference on Computing, Communication and Automation (ICCCA-2021) [\[DOI\]](#)
- **Krishna Mridha** , Aashish Prashad Pandey, Akshay Ranpariya, ”**Web Based Brain Tumor Detection using Neural Network**” , 6th IEEE International Conference on Computing, Communication andz Automation (ICCCA-2021) [\[DOI\]](#)
- **Krishna Mridha** , Aashish Prashad Pandey, Smit Kumbhani, P. Damodharan, ”**Automatic Detect the coronavirus (COVID-19) disease using Chest X-ray and CT images**” , 6th IEEE International Conference on Computing, Communication and Automation (ICCCA-2021) [\[DOI\]](#)
- **Krishna Mridha** , Shah Md. Abul Hasan”**Artificial Intelligence (AI) for Agricultural Sector**” accepted by International Conference on Control, Automation, Power and Singnal Processing (CAPS - 2021) [\[DOI\]](#)
- **Krishna Mridha** , Shakil Sarkar, Dinesh Kumar, ”**Respiratory Disease Classification by CNN using MFCC**” , 6th IEEE International Conference on Computing, Communication and Automation (ICCCA-2021) [\[DOI\]](#)

BOOK CHAPTER PUBLICATIONS

- **Krishna Mridha** ,Fitsum Getachew Tola, Shakil Sarkar, Nazmul Arefin, Sandesh Ghimire, Anmol Aran, Aashish Prashad Pandey ”**Automating Malaria Diagnosis with XAI: Using Deep-Learning Technologies for More Accurate, Efficient, and Transparent Results**” ”International Conference on Multi-disciplinary Trends in Artificial Intelligence”, 2023 [\[Book Chapter \(DOI\)\]](#)
- **Krishna Mridha** , Rabindra Nath Shaw, Dinesh Kumar, ”**Driver Drowsiness Alert System using Real-Time Detection**” , ”Studies in Computational Intelligence”, 2022 [\[Book Chapter \(DOI\)\]](#)
- **Krishna Mridha** ,Sourav Simanta, ”**Diabetes Prediction Using Machine Learning Techniques**”, IEEE IAS Global Conference on Emerging Technologies (GlobConET), 2022 [\[Book Chapter \(DOI\)\]](#)
- **Krishna Mridha** , Suman Jha, Bikash Shah ”**Machine Learning Algorithms for Predicting the Graduation Admission**”, Innovations in Electrical and Electronic Engineering”, ”International Conference on Electrical and Electronics Engineering”, 2022 [\[Book Chapter \(DOI\)\]](#)
- **Krishna Mridha** , Shakil Sarker, ”**Machine Learning in Bioinformatics: New Technique for DNA Sequencing Classification**”, IEEE IAS Global Conference on Emerging Technologies (GlobConET), 2022 [\[Book Chapter \(DOI\)\]](#)

SUMMER SCHOOL OFFERS

- July 22-26, 2024 *Summer School on AI Technologies for Trust, Interoperability, Autonomy and Resilience in Industry 4.0. [Brazil and Spain]*
- August 5-9, 2024 *OSG Summer School [Madison, Wisconsin, USA]*
- September 2-6, 2024 *Applied Harmonic Analysis and Machine Learning [Genova]*

EDITORIAL BOARD MEMBER AND REVIEWER

- 2023 *Reviewer for IEEE 11th Region 10 Humanitarian Technology Conference (R10-HTC)*
- 2023 *Reviewer for 20th India Council International Conference (INDICON)*

PROFESSIONAL MEMBERSHIP

- Member in *The American Health Information Management Association (AHIMA)*, ID: 2989536
- Member in the *Institute of Electrical and Electronics Engineers (IEEE)* , ID: 100090583
- Member in the *American Medical Informatics Association (AMIA)*

AWARDS, ACHIEVEMENTS AND SCHOLARSHIPS

- Awarded a *PhD full Scholarship from Case Western Reserve University* in Biomedical and Health Informatics from August 2023 to June - 2028.
- Received the *Student Volunteer Award* at AMIA 2025, which waived the registration fees. (*Only 5 students selected nationwide*)
- Selected as *Springer Best Research Paper Award* from ICEEE - 2023
- Selected as *IEEE Best Research Paper Award* from ICDCECE - 2023
- Selected as *IEEE Best Research Paper Award* from ICCCA - 2021.
- Selected as *IEEE Best Research Paper Award* from GUCON - 2021.
- Awarded the *Principal's list scholarship for achieving the top position* among all of department - (2014-2017)
- Selected for *Indian Government Full Scholarship* for Pursuing B.Tech in Computer Engineering - (2019-2023)

REFERENCES

Dr. Ming Wang

Professor (with Tenure), Dept. of Population and Quantitative and Health Sciences
Case Western Reserve University
E-mail: mxw827@case.edu

Dr. Lijun Zhang

Associate Professor, Dept. of Population and Quantitative and Health Sciences
Case Western Reserve University
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Dr. Yu Yin

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Case Western Reserve University
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Dr. Madhu Shukla

Associate Professor & Head, Dept. of Computer Engineering - Artificial Intelligence & BIG DATA

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